

Question Bank Mobile Application Dev Module - II

- Explain how app structure in Flutter should be planned to support smooth navigation and routing between multiple screens. Include the role of routes in organizing the app and improving user experience.
- Describe the working of Navigator 1.0 methods and named Navigation in Flutter. Compare these methods and explain how data can be passed between screens while navigating.
- What is named routing in Flutter, and how are route arguments used with it? Explain the process of sending and receiving arguments between screens with a suitable example.
- Explain state management in Flutter and compare Stateless widgets with Stateful widgets. Discuss their features, differences, and suitable use cases in app development.
- Describe how local state is managed in Flutter using `setState()`. Explain its working process, advantages, limitations, and situations where it is most effective.
- What is the Provider package in Flutter? Explain its simplified working mechanism, benefits, and how it helps manage shared state efficiently in applications.
- Discuss the basics of Riverpod in Flutter. Explain its key features, common use cases, and how it improves state management compared to traditional approaches.
- Give an overview of the BLoC pattern and introduce GetX for state and route management in Flutter. Explain their core concepts, advantages, and practical use cases.
- Explain how Git and GitHub are used for collaboration in software development. Discuss the roles of repositories, branching, commits and version control in team projects.
- Describe API integration in Flutter using the `http` package. Explain the steps involved in sending requests, receiving responses, parsing JSON data, and handling errors.
- What is Firebase in Flutter development? Explain the use of Firestore, Realtime Database, and Cloud Storage, highlighting their features and differences.
- Discuss Firebase Authentication in Flutter. Explain the methods of user login and registration, common authentication providers, and its importance in secure app development.
- Explain the use of `print()` statements and the Debug Console in Flutter debugging. Discuss how they help identify errors, track variable values, and monitor application flow during development.
- What is unit testing in Flutter using the `test` package? Explain its purpose, process, and how it is used to verify the correctness of individual functions or logic components.
- Describe Widget Testing and Integration Testing in Flutter. Compare both testing methods and explain their importance in ensuring proper UI behavior and complete app functionality.
- Discuss performance profiling in Flutter using DevTools. Explain how DevTools helps analyze memory usage, CPU performance, rendering speed, and optimize application efficiency.